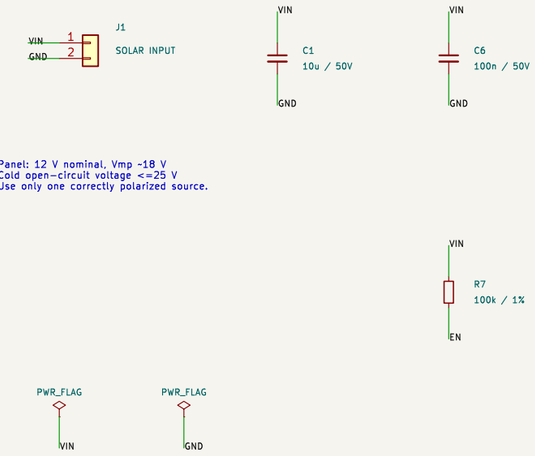
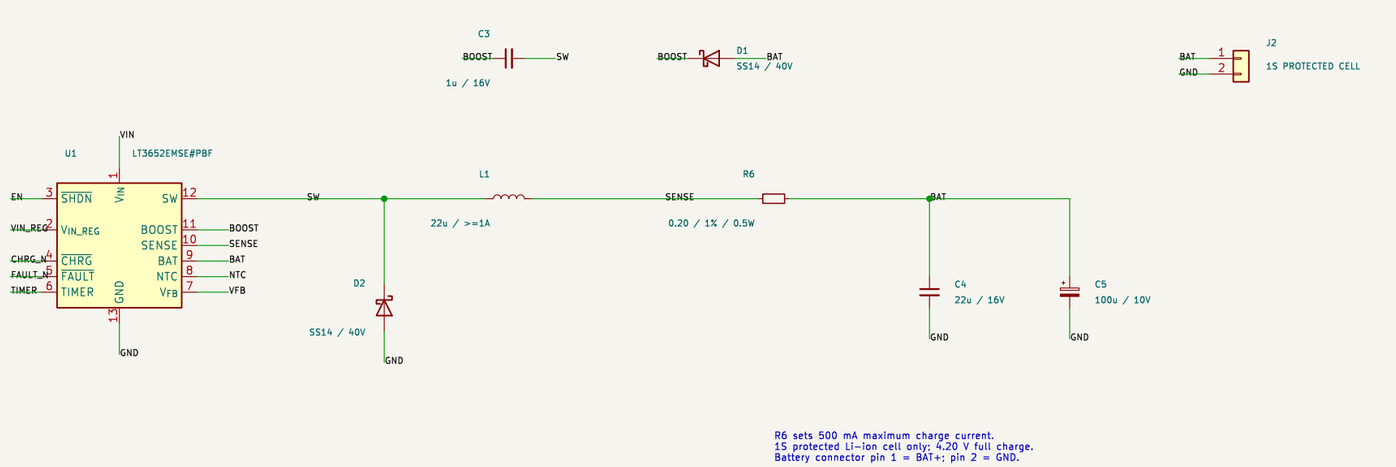
1. SOLAR INPUT / ENABLE



Panel: 12 V nominal, Vmp ~18 V
Cold open-circuit voltage <=25 V
Use only one correctly polarized source.

2. LT3652 CHARGER / POWER STAGE



R6 sets 500 mA maximum charge current.
1S protected Li-Ion cell only: 4.20 V full charge.
Battery connector pin 1 = BAT+; pin 2 = GND.

3. PANEL VOLTAGE REGULATION

VIN_REG = 2.7 V at VIN = 17.874 V.
Fixed panel-voltage tracking; not dynamic MPPT.

4. TIMER



Nominal 3 h; bad-battery preconditioning timeout ~22.5 min.
Timer retries if float not reached.

5. PRECISION BATTERY FEEDBACK

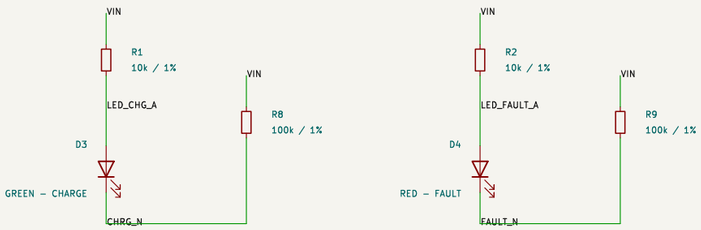
Vfloat = 3.3*(1+60.0k/220k) = 4.200 V
R3 + (R4 || R5) = 250.14 kohm
Use specified 0.1% feedback resistors.

6. TEMPERATURE SENSOR



Attach 10k NTC, B=3380 K, to cell.
Nominal charge window -0 to 40 C.
Open sensor DISABLES temperature monitoring.

7. ADDED STATUS LEDs



Green only: charging >=C/10 (50 mA nominal).
Red only: bad-battery timeout. Both: NTC fault.
Both off: standby, no input, or low charging current.
LEDs are powered from solar VIN; off does not prove full charge.



VIN-LEVEL STATUS
NOT MCU SAFE
SET = VIN_REG sense only

ENGINEERING PROTOTYPE – ERC/DRC do not replace bench verification. No lithium cell until dummy-load checks pass.

Based on the uploaded Fernando S.M.H.G. project; corrected circuit and layout
1S Li-ion / 4.20 V / 500 mA – engineering prototype

Sheet: /		
File: Solar_Charger_LED_RevA.kicad_sch		
Title: Solar Battery Charger – Charging and Fault LEDs		
Size: A3	Date: 2026-11-05	**Rev: A**
KiCad E.D.A. 9.0.2		Id: 1/1